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Summary

Senior AI/ML Engineer & Platform Architect — 14+ years delivering scalable production systems to enterprise customers across collaboration, observability, and healthcare. Currently leads the AI layer of Cisco's AssessmentX. Combines hands-on engineering depth (fine-tuning models, agentic systems, retrieval) with the architect's craft of translating customer outcomes into engineering deliverables, operating across engineering teams, internal stakeholders, and enterprise customer leadership. **M.S. Computer Science (Artificial Intelligence), Georgia Tech, Dec 2024.**

Applied ML Research

Domain-Specialized SLM Fine-Tuning — Llama 3.2 3B QLoRA for US Community-Healthcare Q&A. Built an end-to-end, fully local pipeline to download, ingest, and process ~3,200 publicly-available **Community Health Needs Assessment (CHNA) reports** from US hospitals across all 50 states and territories, then fine-tuned a Llama 3.2 3B Instruct base into a CHNA-domain Q&A specialist answering complex questions about healthcare infrastructure, resource gaps, and demographic needs. *No proprietary APIs in data prep, training, or evaluation.* [HuggingFace](#) · [GitHub](#) · [Methodology write-up](#).

- **Training architecture.** LoRA rank 32 across attention + MLP; 4-bit NF4 + double-quant; paged AdamW 8-bit; gradient checkpointing; NEFTune embedding noise; single consumer GPU (RTX 4090, 24 GB). Synthetic Q&A generated by a local Ollama teacher (Llama 3.1 8B Q4) with faithfulness filtering.
- **Hospital-level holdout splits.** Train / val / test partitioned by hospital, not by row — a hospital appearing in train cannot leak into eval, preventing the model from memorizing near-identical annual CHNAs.
- **Multi-metric evaluation.** Perplexity, pairwise judging vs base (Qwen 2.5 7B as the LLM-as-judge, different family from teacher to reduce shared bias), 1–5 rubric, ROUGE-L, and faithfulness on the held-out test set. Pairwise win-rate vs base Llama 3.2 3B Instruct = **68%** (n=300); rubric avg = **4.0 / 5.0** (n=500); faithfulness = **0.78**.

Experience

Cisco Systems

CX Healthcare | AppDynamics | CX Collaboration

Nov 2018 – Present

San Francisco Bay Area, CA

Software Architect — AssessmentX AI Platform

Nov 2024 – Present

Lead architect of AssessmentX AI — a FastAPI companion service to AssessmentX's Spring-Cloud microservices, acting as a one-stop adapter for ingestion, retrieval, search, and RAG-augmented context-chat (conversation + on-demand deep analysis). Consumed via APIs by AssessmentX, also shippable standalone with its own Angular UI. Each bullet below is a capability of this one platform.

- **Agentic SDK.** Domain-agnostic, LangGraph-based SDK so new agents ship by writing only a skill manifest and tools implementation — 5 graph templates, three-tier model resolution, Send-based parallel orchestration, Pydantic structured outputs, and bounded retry are reused across agents. Three production specialist agents shipped on it; custom graphs supported when templates don't fit. Built so AssessmentX and any future Cisco product stand up new agents on shared rails.
- **Hybrid retrieval & NL→MongoDB query generation.** Rebuilt the platform's ingestion and retrieval as a 7-phase pipeline across six document formats; hybrid retrieval (vector + BM25 + Reciprocal Rank Fusion + LLM re-rank) on MongoDB Atlas Vector Search. Includes a router / planner chat graph that dispatches structured-data intents to LLM-generated MongoDB aggregations and reserves vector RAG for unstructured questions; provider-agnostic across Cisco BridgeIT.
- **Debugger Studio.** LangSmith-grade, self-hosted observability for the platform's LangGraph agents — built because regulated customer data flows through every prompt, retrieval, and agent trace and HIPAA-aware observability has to stay inside Cisco's network. Auto-discovers agents in the codebase; live DAG, span tree, per-node input and LLM-response inspection, latency waterfall, token and cost accounting, LLM-as-judge scoring, run comparison, and an in-browser prompt playground with from-run replay.
- **Governed LLM codegen (Assessment Studio module).** Same parse → score → render pipeline AssessmentX runs in production, generated on demand: an LLM authors the per-pack Python from a markdown spec, gated by an AST allowlist, then human approval (SHA-256 over the script), then a frozen-pack snapshot; approved scripts execute in a subprocess sandbox with memory and time caps.
- **Agentic SDLC framework.** Multi-IDE (Cursor, Claude Code, GitHub Copilot, Windsurf, VSCode AI) developer toolkit distributed by git subtree as ~55 skills (Jira / Confluence / SharePoint / Splunk / OpenShift), an age-encrypted secret vault, a TypeScript MCP server (31 tools / 10 resources / 5 prompts), and security gates on CSDL, Project CodeGuard (OWASP), Trivy, Semgrep, OWASP ZAP, Anchore, and Cisco AI Defense MCP scanning.

Software Consulting Engineer IV — CX Healthcare, AssessmentX

Sep 2022 – Nov 2024

*Built Cisco's internal assessment-automation platform **AssessmentX** from the ground up with three other engineers, in support of the CX Healthcare PS-delivery team running HIMSS healthcare assessments for hospitals, providers, and life-sciences customers.*

- **Platform build — backend, AI service, and frontend.** Architected and shipped end-to-end: Java 21 / Spring Boot 3.5 / Spring Cloud 2025 Maven monorepo of 10 microservices + Python AI companion (Flask, later ported to FastAPI) + Angular 18 standalone UI, wired by Feign over Netflix Eureka; multi-stage Docker on hardened Alpine JRE 21 to Cisco ECR. Internal customer (Cisco PS delivery) submits Excel questionnaires; certified practitioners drive the workflow (parse → weighted scoring → DOCX/PPTX deliverables).
- **Healthcare & capability assessment suite.** Led 4-lane parallel delivery of a new assessment type end-to-end (questionnaire ingestion + data-collection UI, weighted scoring + reporting dashboard, DOCX/PPTX deliverables, practitioner workshop tab). Then rationalized the suite into a shared questionnaire component and score model across five phases — now the template every new assessment type follows.

- **Cross-stack integration & security hardening.** Unified project lifecycle and document push across the Java platform and the AI companion via Feign-proxied per-project AI flags; consolidated secrets into AWS Secrets Manager; ported the AI companion from Flask to FastAPI. Drove ongoing CSDL and SonarQube compliance — Spring / spring-security / bouncy-castle upgrades, OWASP ZAP, OAuth + CSP hardening.

Software Consulting Engineer IV — AppDynamics, Customer Engineering

Feb 2022 – Sep 2022

Internal team transfer from CX Collaboration to AppDynamics — customer-facing engineering-consulting role on observability strategy for enterprise customers.

- **Technical SME, customer engagement lifecycle.** Embedded with enterprise AppDynamics customers as their architecture-level technical subject-matter expert on observability. Walked through the customer's architecture, codified key performance indicators with their engineering leadership, and engaged with the C-suite to capture the observability vision; translated it into engineering requirements, verified with the customer's technical teams, and stayed engaged through delivery — including building custom AppDynamics plugins when needed. Customer roster: Tech Mahindra, Fastenal, Carnival, RND. Earned a top-quartile bonus and a formal recommendation from the engineering lead.

Software Consulting Engineer III → IV — CX Collaboration, Cross Domain Center of Excellence

Nov 2018 – Feb 2022

Engineering lead on the Collaboration Workflow Manager (CWM) — Cisco CDCoE's enterprise platform for onboarding and provisioning users onto Cisco collaboration infrastructure for headline customers including Bank of America and Apple. Promoted from SCE III to SCE IV in Apr 2021.

- **CWM architecture lead.** Owned the data-extraction and data-aggregation microservices; drove cluster consolidation and data-migration services. Led customer rollouts for Bank of America (CDCoE's headline financial-services customer of the period) and Apple (collaboration-infrastructure provisioning).
- **Webex Contact Center agent-statistics dashboard.** Designed and shipped end-to-end and independently (Angular + Spring Cloud + MongoDB) — live agent, queue, and contact metrics for Webex CC customers, complete-stack ownership including UX.

Earlier: Infosys Limited (Aug 2011 – Nov 2018, Bangalore → SF Bay Area)

- **Backend / Systems Engineer, JPMorgan Chase provisioning platform** (Bangalore, 2011–2017). Built Java + jBPM workflow scripts and Spring REST + SOAP-WS northbound APIs for JPMorgan Chase's enterprise provisioning platform — the Infosys analog to what Cisco CDCoE later built as CWM. Processed 500K+ requests across a 5-year engagement. Promoted twice; relocated to the US in 2017.
- **Full-Stack engineer, Custom Application Development & Integrations** (CADI, SF Bay Area, 2017–2018). Continued the JPMorgan Chase engagement from the US delivery center; owned 100% of usability improvements; delivered a complete UI redesign and the platform's log-management system.

Education

M.S. Computer Science (Artificial Intelligence), Georgia Institute of Technology — Atlanta, GA (Online)

Dec 2024

B.Tech. Electronics & Communications, Punjabi University, Patiala — Punjab, India

Jan 2011

Technical Skills

AI / ML Skills: Fine-tuning SLMs / LLMs · agentic SDK development · agent development · LLM codegen (AST-gated) · custom agentic observability · Model Context Protocol (MCP) · multi-agent orchestration · type-agnostic Document IR ingestion · structured & unstructured data retrieval · intent classification · tokenization · embeddings (Matryoshka · OpenAI · Google) · hybrid search (cosine similarity / dot product + BM25 / TF-IDF, variable weights) · LLM re-ranking · BGE re-ranker · Reciprocal Rank Fusion · NL→MongoDB / NL→SQL querying · GraphRAG · hierarchical retrieval · structured document topology-based retrieval · LLM-as-judge · pairwise evaluation · AI guardrails · prompt-injection defense · HIPAA-aware AI observability · LoRA / QLoRA · synthetic data generation

AI / ML Tools: LangChain · LangGraph · LangSmith · RAGAS · PyTorch · HuggingFace Transformers · TRL · PEFT · pandas · NumPy · scikit-learn · XGBoost · TensorFlow · AWS Bedrock · AWS SageMaker · GPT-4o / Anthropic Claude / Gemini / Llama / Azure OpenAI

Backend Engineering & Databases: Python (FastAPI · Flask) · Java 21 · Spring Boot 3 / Spring Cloud · Maven · REST / gRPC / SSE · microservices · distributed systems · event-driven architecture · system design · MongoDB Atlas · PostgreSQL · DuckDB

Frontend: Angular 18 · TypeScript · standalone components · signals · Figma AI designs

Security: HIPAA-aware design · FedRAMP-adjacent · CSDL compliance · OWASP ZAP · Trivy · Semgrep · Anchore · OAuth · JWT · AWS Secrets Manager · age-encrypted secret vault

Development Tools: Cursor · Claude Code · GitHub Copilot · Windsurf · VSCode AI · Conda · Docker · Kubernetes · OpenShift · Terraform · CI/CD · Jenkins · AWS (S3 · ECR · Secrets Manager · OpenSearch · EC2 / IAM) · GCP · Azure · Jira · Confluence · SharePoint · Splunk · author of a ~55-skill agentic SDLC framework + TypeScript MCP server (31 tools / 10 resources / 5 prompts)